INCH-POUND

MIL-B-49461/2B(ER)

3 July 1989

SUPERSEDING
MIL-B-49461/2A(ER)
10 July 1987

MILITARY SPECIFICATION SHEET

BATTERY, NON-RECHARGEABLE, LITHIUM THIONYL CHLORIDE, BA-6598/U

This specification is approved for use by US Army Laboratory Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-B-49461(ER).

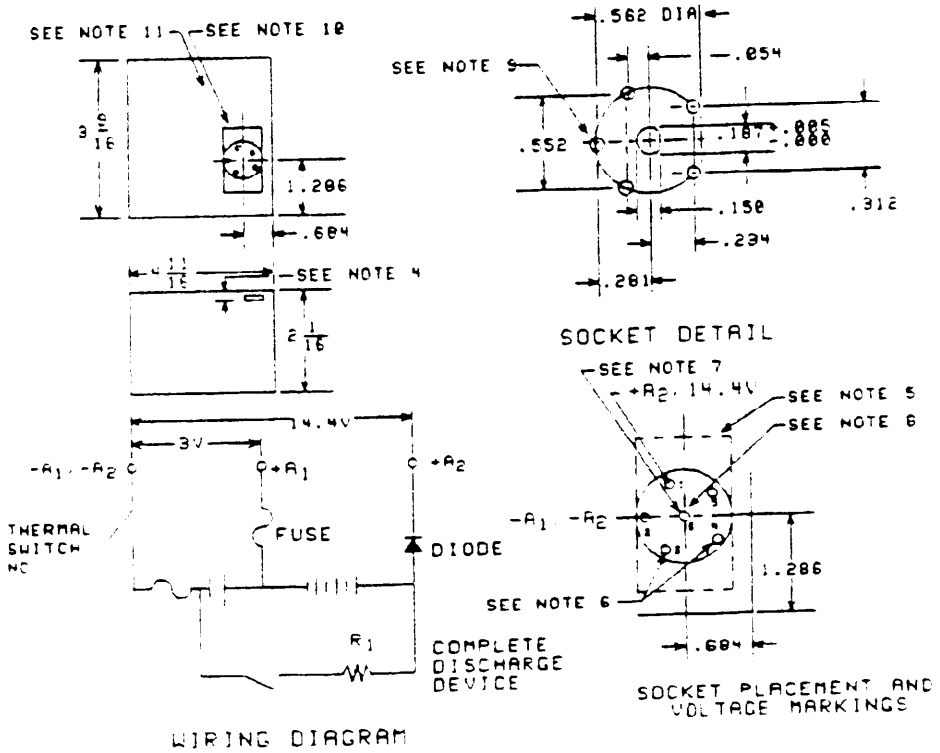


Figure 1. Battery dimensions and wiring schematic.

FSC 6135

AMSC N/A
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NOTES

1. All dimensions are in inches.
2. Tolerances, unless otherwise specified, are ±0.005 inch for decimals.
3. Tolerances of length, width, and height of battery are ±1/16 inch for fractions.
4. The minimum depth of socket well shall be 1/2 inch.
5. The size of the insulating plate is optional.
6. Socket insert contact not required.
7. Pin-circle center.
8. "D" hole for plug positioning.
9. Five holes to fit 1/8 inch pins, 7/16 inch in length.
10. Socket must float 3/32, + 1/32, 1/16 inch in all directions from the specified pin circle center location in the plane of the socket from reference dimension.
11. The socket shall be supported and mounted so that the top surface of the socket shall not protrude above the adjacent outside surface of the jacket and shall not be more than 1/16 inch below adjacent outside surface of the jacket when used initially, during, and after subsequent insertions of the mating plug during contractual testing.
12. R_1 value shall be between 125 ohms and 175 ohms and have a power rating of 2.0 watts.
13. Socket wiring schematic to be printed or embossed on jacket in suitable location near the socket on the terminal face of the battery.

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REQUIREMENTS:

Dimensions and configuration: See Figure 1.

	A_1 Unit (volts)	A_2 Unit (volts)
Maximum Voltage	3.70	14.80
Minimum Voltage	2.50	10.00

A 2.25 ampere slow blow non-replaceable fuse shall be used in both the negative and the $+A_1$ legs.

A diode shall be wired in series with the A2 section. See paragraph 3.5.5.2 of the basic specification.

Terminals: Five (5) hole socket type with no obstruction of any holes.

Weight (Maximum): 1.7 pounds (772 grams)

Capacity tests: When the battery is tested in accordance with the methods of examination and tests of this specification, the minimum capacity test requirements shall be as specified below:

Capacity Test	Battery Capacity in hours to 10.00 Volts
I	7.0
L	3.5
H	13.0
HT	12.6
LT	3.0
T	7.0
D	7.0
ILR	1500
IT	6.3

Initial Voltage Delay: When the battery is subjected to the I, IT, H, HT, T, D and ILR capacity test, the time required at the beginning of discharge of the battery to reach a voltage of 10.00 volts after the specified load is applied shall not be more than 5.0 seconds. For the L and LT capacity tests the time to reach a voltage of 10.00 volts after the load is applied shall not be more than 60 seconds.

High temperature switch. A high temperature switch or thermal fuse shall be employed internal to the battery in the negative leg as described in paragraphs 3.5.5 and 3.5.5.1 of the basic specification.

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Complete discharge device: This device shall be inserted in the battery as shown on the circuit schematic and shall have the characteristics as described in para 3.5.5.3 of the basic specification.

METHODS OF EXAMINATION AND TESTS:

(1) Storage: Details on storage conditions for all capacity tests are specified in basic specification.

(2) Discharge:

Cells: Discharge at 2.0 amps constant current at 70±5° F for a minimum of 6.22 hours to 2.50 volts cut-off. Continue discharge to zero volts. Then discharge at 1.0 ampere constant current for 7.0 hours into voltage reversal.

Batteries: The A_2 unit ($-A_1$, $-A_2$ to $+A_2$) shall be discharged as follows:

For all capacity tests except the H, HT, and ILR, the batteries shall be discharged at two amperes constant current to zero volts. For the H and HT capacity test the batteries shall be force discharged at two ampere constant current for five minutes followed by open circuit for five minutes. This cycle shall be repeated until the closed circuit voltage reaches zero volts. For the I capacity test the batteries shall be force discharged at one ampere constant current for 7.2 hours immediately after zero volts is obtained. There shall be no venting, fire or explosion when the batteries are force discharged. For the ILR test a fixed resistance of 2210 ohms shall be used to a cut-off voltage of 10.0 volts or 1500 hours.

Closed circuit voltage: Closed circuit voltage measurements shall be made with a direct current voltmeter of proper range and sensitivity (see basic specification) using load resistances as shown in the following table:

	UNIT RESISTANCE	MINIMUM PERMISSIBLE VOLTAGE
A_1	8.0 ohms, ±1%	2.4 volts
A_1	14.2 ohms, ±1%	10.0 volts

Voltage shall be above "MINIMUM PERMISSABLE VOLTAGE" within five (5) seconds after load is applied.

Socket Strength Test (First Article only): Before conducting the jacket integrity and visual mechanical examinations specified in Table III, the 5 batteries shall be subjected to the following:

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Using a 0.250 ±0.050 inch diam. rod, apply a force to the center of the socket and deflect it downward a distance of 0.250, + 0.0625, -0 inch below the top surface of the battery.

Battery Label (only for this battery):

"DO NOT CHARGE, SHORT CIRCUIT, INCINERATE, MUTILATE OR TEST WITH U-410 ADAPTOR OTHERWISE BATTERY MAY VENT OR RUPTURE RELEASING TOXIC MATERIALS"

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodian:
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